

ReviewQuestions_SE_Only

Chapter 1: Introduction

- **What are the essential attributes of good software? ***
中文：好的软件应具备哪些基本属性？
 - **What are the two fundamental types of software product?**
中文：有哪两种基本类型的软件产品？
 - **What is software engineering?**
中文：什么是软件工程？
 - **What are the four fundamental activities in software processes?**
中文：软件过程中的四个基本活动是什么？
 - **What is the distinction between computer science and software engineering?**
中文：计算机科学与软件工程之间的区别是什么？
 - **What are the 3 general issues that affect many different types of software? ***
中文：会影响许多不同类型软件三个常见问题是什么？
 - **List 5 different types of software application. ***
中文：列举5种不同类型的软件应用。
 - **What software engineering fundamentals apply to all types of software systems? ****
中文：哪些软件工程基本原则适用于所有类型的软件系统？
 - **What are three key characteristics of the engineering of web-based software engineering? ***
中文：网络软件工程的三个关键特征是什么？
 - **What is a software engineering code of ethics?**
中文：什么是软件工程道德准则？
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Chapter 2: Software processes

- **What are the fundamental activities that are common to all software processes?**
中文：所有软件过程共同具有的基本活动是什么？
- **List the 3 generic process models that are used in software engineering?**
中文：列出在软件工程中使用的三种通用过程模型。
- **Why are iterations usually limited when the waterfall model is used?**
中文：当使用瀑布模型时，为什么迭代通常是有限制的？
- **What are the three benefits of incremental development, compared to the waterfall model?**
中文：与瀑布模型相比，增量开发的三个优点是什么？

- **What are the development stages in reuse-based development? * ***
中文：重用开发的开发阶段是什么？
 - **What are the principal requirements engineering activities?**
中文：什么是需求工程的主要活动？
 - **Why is it increasingly irrelevant to distinguish between software development and evolution?**
中文：为什么越来越没有必要区分软件开发和演化？
 - **What are the advantages of using incremental development and delivery? * ***
中文：使用增量开发和交付的好处是什么？
 - **What are the 4 sectors in each loop in Boehm's spiral model?**
中文：Boehm的螺旋模型中每个循环中的4个部分是什么？
 - **What are the six fundamental best practices in the RUP(Rational Unified Process)? * ***
中文：RUP（统一软件开发过程）中的六个基本最佳实践是什么？
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Chapter 3: Agile Software Development

- **What are the shared characteristics of different approaches to rapid software development? * ***
中文：快速软件开发方法的不同方法共享哪些特征？
- **For what types of system are agile approaches to development particularly likely to be successful?**
中文：敏捷方法在哪些类型的系统中特别容易成功？
- **List the 5 principles of agile methods. ***
中文：列出敏捷方法的5个原则。
- **List 4 questions that should be asked when deciding whether or not to adopt an agile method of software development.**
中文：决定是否采用敏捷软件开发方法时应提出哪四个问题？
- **What are three important characteristics of extreme programming? ***
中文：极限编程的三个重要特点是什么？
- **What is test-first development?**
中文：什么是测试优先开发？
- **What are the possible problems of test-first development?**
中文：测试优先开发可能存在的问题是什么？
- **Briefly describe the advantages of pair programming.**
中文：简要描述成对编程的优点是什么？
- **What is a Scrum sprint? ***
中文：什么是Scrum开发的冲刺？

- **What are the barriers to introducing agile methods into large companies? ***
中文：在大型公司中引入敏捷方法的障碍是什么？
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Chapter 4: Requirements Engineering

- **What are user requirements and system requirements?**
中文：用户需求和系统需求是什么？
 - **What is the distinction between functional and non-functional requirements?**
中文：功能需求和非功能需求有什么区别？
 - **List 3 types of non-functional requirement. ***
中文：列举三种非功能需求类型。
 - **What is the software requirements document?**
中文：软件需求文档是什么？
 - **What is the distinction between the terms 'shall' and 'should' in a user requirements document, which is written in natural language?**
中文：自然语言用户需求文档中“将”和“应该”的区别是什么？
 - **What are the main advantages of using a standard format to specify requirements?**
中文：使用标准格式规定需求的主要优点是什么？
 - **What are the principal stages of the requirements engineering process?**
中文：需求工程过程的主要阶段是什么？
 - **Give 5 reasons why eliciting requirements is difficult. ***
中文：列举5个引起需求获取困难的原因。
 - **What should be included in a scenario? ***
中文：场景应包含什么？
 - **What is a use-case? ***
中文：什么是用例？
 - **What is ethnography and how is it used in requirements elicitation? ***
中文：什么是人类学，以及如何在需求获取过程中使用它？
 - **What checks should be applied during requirements validation? ***
中文：在需求验证过程中应用哪些检查？
 - **List three requirements validation techniques. ***
中文：列举三种需求验证技术。
 - **What is requirements management? ***
中文：什么是需求管理？
 - **What are the stages in the requirements change management process? ***
中文：需求更改管理过程的阶段是什么？
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Chapter 5: System modeling

- **What perspectives may be used for system modelling? ***
中文：哪些视角可用于系统建模？
 - **What UML diagram types may be used to represent the essential features of a system?**
中文：哪些UML图类型可用于表示系统的基本特征？
 - **What is described in a context model?**
中文：上下文模型中描述了什么？
 - **How are activity diagrams used in describing the context of use of a system?**
中文：活动图在描述系统使用背景时如何使用？
 - **What are the principal components of a textual use-case description? ***
中文：文本用例描述的主要组成部分是什么？
 - **What is illustrated in a UML sequence diagram? ***
中文：UML时序图表示什么？
 - **How is generalization(inheritance) used to simplify the models of a system with many similar objects?**
中文：如何使用泛化（继承）来简化具有许多相似对象的系统的模型？
 - **What is the basic assumption that underlies event-driven modelling?**
中文：事件驱动建模的基本假设是什么？
 - **What are the claimed benefits of model-driven engineering? ***
中文：模型驱动工程的主张的好处是什么？
 - **What are the three types of abstract system model that are recommended by the MDA(Model Driven Architecture) method? ***
中文：什么是MDA（模型驱动架构）方法推荐的三种抽象系统模型类型？
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Chapter 6: Architectural design

- **What are the advantage of explicitly designing and documenting a software architecture?**
中文：明确设计和记录软件架构的优势是什么？
- **What are the two ways in which an architectural model of a system may be used?**
中文：系统的架构模型可能有哪两种用途？
- **List 4 fundamental questions that should be addressed in architectural design. ***
中文：在架构设计中应解决的四个基本问题是什么？
- **What are the fundamental architectural views proposed in Krutchen's 4+1 model?**
中文：Krutchen的4+1模型中提出的基本架构视图是什么？

- **What is an architectural pattern? ***
中文：什么是架构模式？
 - **What is the fundamental characteristic of a repository architecture?**
中文：仓库架构的基本特征是什么？
 - **What is the most important advantage of a client-server architecture?**
中文：客户端-服务器架构的最重要优势是什么？
 - **Briefly describe pipe and filter architecture.**
中文：简要描述管道和过滤器架构。
 - **What are transaction-processing applications? ***
中文：什么是事务处理应用程序？
 - **What are the principal functions of the 4 layers in a generic information system architecture? ***
中文：通用信息系统架构的4层主要功能是什么？
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Chapter 7: Design and implementation

- **What are the 5 key activities in an object-oriented design process? ***
中文：面向对象的设计过程中的五个关键活动是什么？
- **What do you understand by the system context and interaction model?**
中文：系统上下文和交互模型是什么？
- **Briefly describe 3 approaches that may be used to identify object classes? ***
中文：简要描述可用于识别对象类的三种方法？
- **Why is it important to specify the interfaces of components that are being developed by a software engineering team?**
中文：对于由软件工程团队开发的组件进行接口规格说明很重要，为什么？
- **What do Gamma et al. suggest are the four essential elements of a design pattern? ***
中文：Gamma等人建议设计模式的四个基本元素是什么？
- **How do design patterns contribute to reuse? ***
中文：设计模式如何促进重用？
- **What are the 4 levels at which software reuse is possible?**
中文：软件重用有哪四个水平？
- **What are the principal aims of software configuration management? ***
中文：软件配置管理的主要目标是什么？
- **What are essential tools in a software development platform?**
中文：软件开发平台中的基本工具是什么？
- **Briefly describe the idea of open-source development.**
中文：简要描述开源开发的想法。

Chapter 8: Testing

- **What is the distinction between validation and verification?**
中文：验证和验证之间的区别是什么？
- **What are the advantages of inspections over testing? ***
中文：检查比测试有哪些优点？
- **Briefly describe the three principal stages of testing for a commercial software system.**
中文：简要描述商业软件系统的三个主要测试阶段。
- **What tests should be included in object class testing?**
中文：对象类测试中应包括哪些测试？
- **What guidelines does Whittaker suggest for defect testing?**
中文：Whittaker为缺陷测试提出了哪些准则？
- **What is an equivalence partition? Give an example. ***
中文：什么是等价划分？可以举出一个例子吗？
- **What are the three important classes of interface errors?**
中文：接口错误的三个重要类别是什么？
- **What should be the principal concerns of system testing?**
中文：系统测试应该关注哪些主要方面？
- **Briefly summarize the test-driven development process.**
中文：简要概述测试驱动型开发过程。
- **What is scenario testing?**
中文：什么是场景测试？
- **What is stress testing and why is it useful?**
中文：什么是压力测试，为什么有用？
- **What are the three types of user testing?**
中文：用户测试有哪三种类型？

Chapter 9: Evolution

- **Why is software evolution important?**
中文：软件演进为什么很重要？
- **What are the stages in the system evolution process and what triggers that process? ***
中文：系统演进过程的阶段是什么，以及什么触发了该过程？
changes of requests triggers the evolution

- impact analysis
 - release planning
 - change implementation
 - system release
 - **Why might it sometimes be necessary to bypass the normal change management system and make urgent changes to a system?**
中文：为什么有时需要绕过正常的变更管理系统并对系统进行紧急变更？
 - **What are Lehman's Laws and how were they derived? ***
中文：什么是Lehman定律，它们是如何推导出来的？
 - **What are the three different types of software maintenance and how is effort distributed across these maintenance types?**
中文：软件维护的三种不同类型是什么，这些维护类型的工作量分配如何？
 - **What factors should be assessed to understand the relationship between a system and its environment? ***
中文：应该评估哪些因素以了解系统与其环境之间的关系？
 - **What process metrics might be used to assess maintainability? ***
中文：哪些过程指标可用于评估可维护性？
 - **What are the principal systems re-engineering activities? ***
中文：主要的系统再造活动有哪些？
 - **What are the strategic options for legacy system evolution?**
中文：传统系统演进的战略选项是什么？
 - **List four important factors used to assess applications for evolution.**
中文：列出用于评估应用程序演进的四个重要因素。
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Chapter 10: Sociotechnical systems

- **What is the difference between the business process layer and the organizational layer in the sociotechnical systems stack.**
中文：社会技术系统堆栈中的业务流程层和组织层之间有什么区别？
- **What is the difference between a technical and a sociotechnical system?**
中文：技术系统和社会技术系统之间有什么区别？
- **What are emergent properties?**
中文：什么是涌现属性？
- **What are three influences on the reliability of a system?**
中文：影响系统可靠性的三个因素是什么？
- **Why are sociotechnical systems non-deterministic?**
中文：为什么社会技术系统是非确定性的？

- **What is a wicked problem?**
中文：什么是棘手的问题？
 - **What are the three principal stages of systems engineering? ***
中文：系统工程的三个主要阶段是什么？
 - **What are the main drivers for system procurement decisions? ***
中文：系统采购决策的主要驱动因素是什么？
 - **Why are plan-driven(rather than agile) processes used in systems engineering?**
中文：为什么在系统工程中使用计划驱动（而不是敏捷）流程？
 - **What are latent conditions and active failures?**
中文：什么是潜在条件和主动故障？
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Chapter 11: Dependability and Security

- **Give three reasons why a system's dependability is more important than its detailed functionality.**
中文：给出系统的可靠性比其详细功能更重要的三个原因。
- **What are the four principal dependability properties?**
中文：四个主要的可靠性属性是什么？
- **List two other system properties that are sometimes considered to be dependability properties.**
中文：列出另外两个有时被认为是可靠性属性的系统属性。
- **Briefly define what availability means?**
中文：简要定义可用性的含义？
- **Explain how a relatively unreliable system can provide a high level of availability.**
中文：解释一个相对不可靠的系统如何提供高水平的可用性。
- **Explain the difference between a system fault and a system failure.**
中文：解释系统错误和系统故障之间的区别。
- **What is the most important difference between the two classes of safety-critical system?**
中文：这两类安全关键系统之间最重要的区别是什么？
- **What is the distinction between a hazard and an accident?**
中文：危险和事故有什么区别？
- **What are the three principal threats to the security of a system? ***
中文：系统安全的三大主要威胁是什么？
- **What are the three controls may be put in place to enhance system security? ***
中文：可以实施哪些三个控制措施来增强系统安全性？